EE2001 Electrical Circuits-I:

Continuation to:

Techniques of Circuit Analysis

30 October 2023

Outline

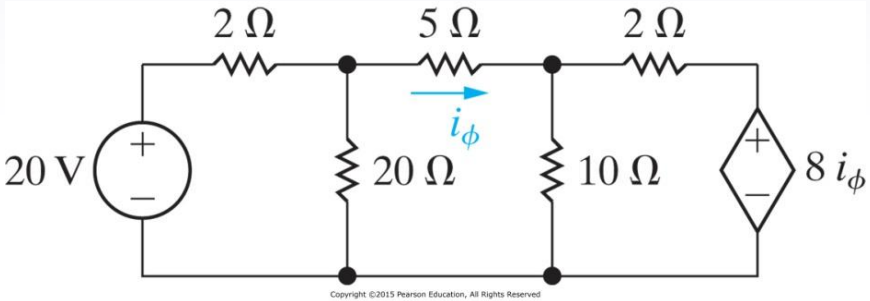


1. **The Node-Voltage Method and Dependent Sources**
2. **The Node-Voltage Method: Some Special Cases**
3. **The Mesh-Current Method and Dependent Sources**
4. **The Mesh-Current Method: Some Special Cases**
5. **The Node-Voltage Method Versus The Mesh-Current Method**
6. **Source Transformations**

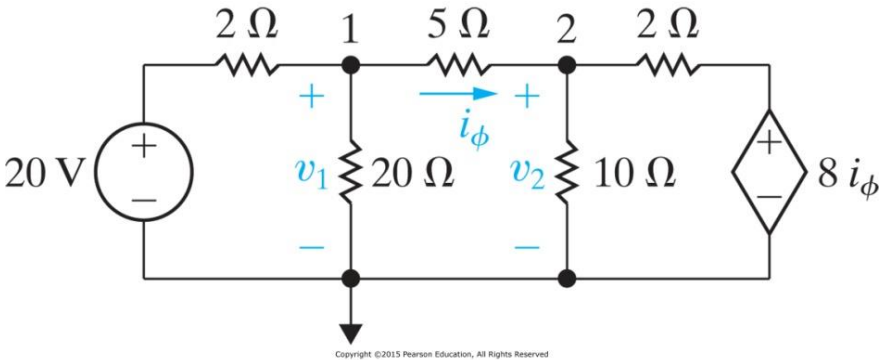
To be continued...

The Node-Voltage Method and Dependent Sources

- In case there are dependent sources in the circuit, the constraint equations for these sources need to be considered as well.
- Example:



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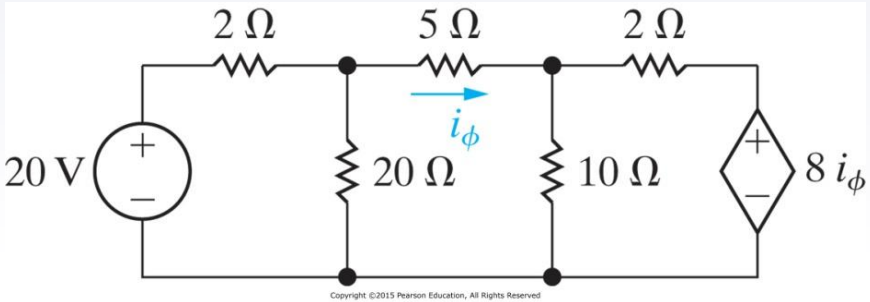


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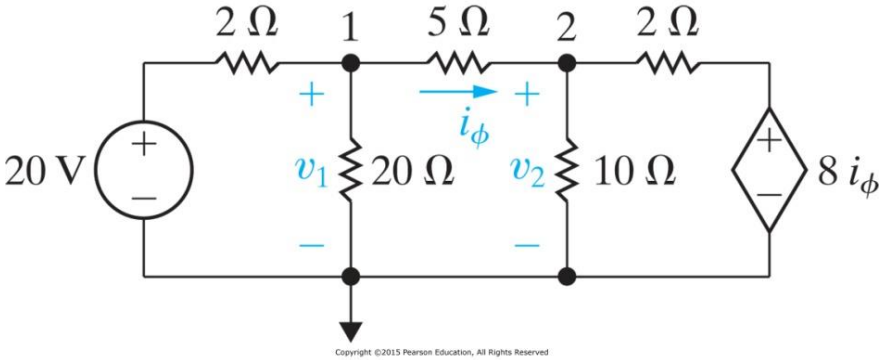
- The circuit has three eseential nodes. Hence, we need two node-voltage equations to describe the circuit.
- Four branches terminate on the lower node, so we select it as the reference node.
- Two unknown node voltages are defined on the circuit.

The Node-Voltage Method and Dependent Sources

- Example:



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- Summing the currents away from node 1 generates the equation:

$$\frac{v_1 - 20}{2} + \frac{v_1}{20} + \frac{v_1 - v_2}{5} = 0$$

- Summing the currents away from node 2 yields:

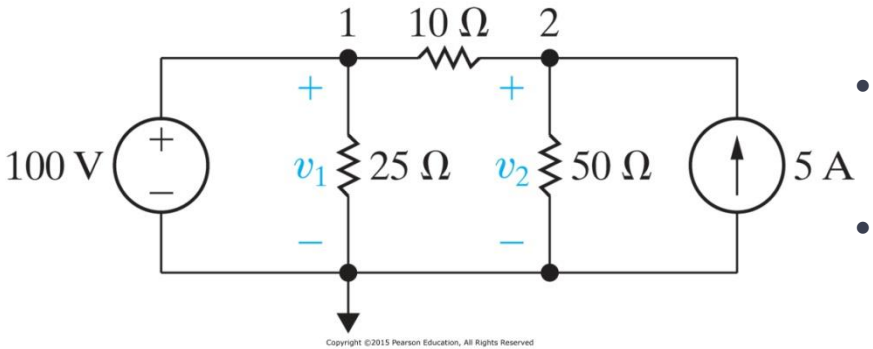
$$\frac{v_2 - v_1}{5} + \frac{v_2}{10} + \frac{v_2 - 8i_\phi}{2} = 0$$

- To eliminate i_ϕ, we must express this controlling current in terms of node voltages:

$$i_\phi = \frac{v_1 - v_2}{5}$$

The Node-Voltage Method: Some Special Cases

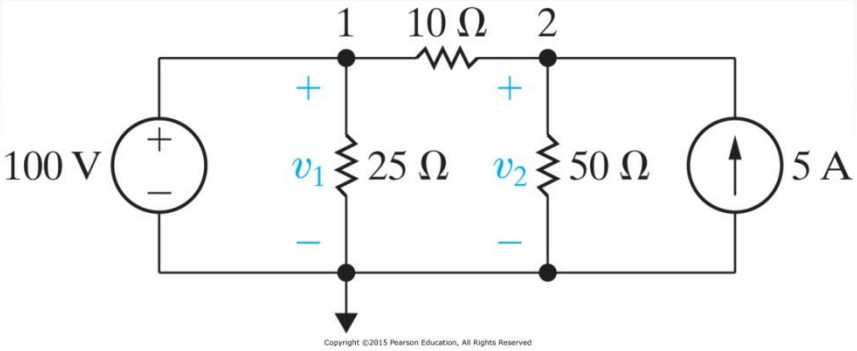
- When a voltage source is the only element between two essential nodes, the node-voltage equation method is simplified. The number of unkown node voltages and therefore the number of equations are reduced.
- Example:



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- There are three essential nodes in the given circuit, so two simultaneous equations are needed.
- However, 100 V source constraints the voltage between node 1 and the reference node to 100 V.
- This means that there is only one unknown node voltage (v_2). Solution of this circuit involves only one node-voltage equation at node 2.

The Node-Voltage Method: Some Special Cases



- Node-voltage equation for node 2:

$$\frac{v_2 - v_1}{10} + \frac{v_2}{50} - 5 = 0$$

$$v_1 = 100\ \text{V}$$

$$\frac{v_2 - 100}{10} + \frac{v_2}{50} - 5 = 0$$

$$\frac{5v_2 - 500}{50} + \frac{v_2}{50} = 5$$

$$5v_2 - 500 + v_2 = 250$$

$$6v_2 = 750$$

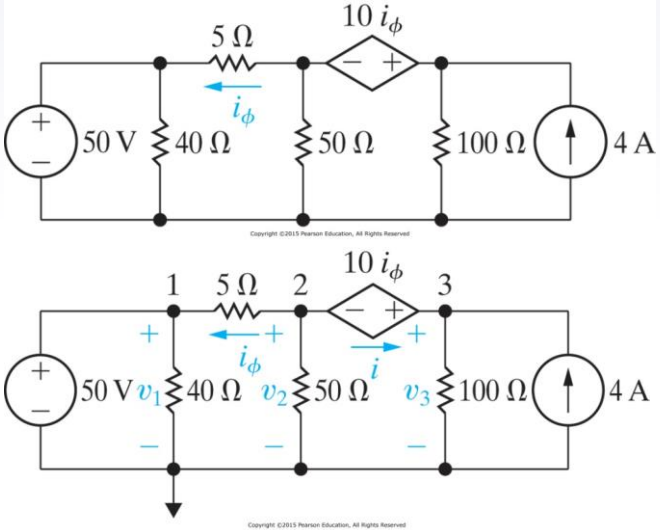
$$v_2 = 125\ V$$

- The current in every branch can be easily calculated based on the found voltage value.

The Node-Voltage Method: Some Special Cases

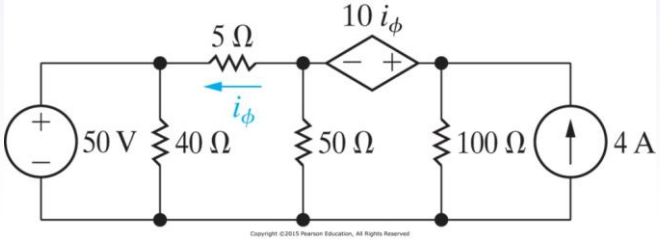
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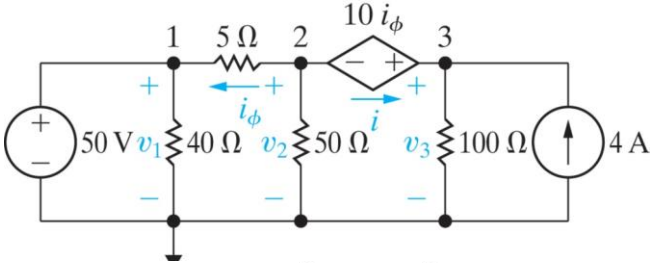
- When a voltage source is the only element between two essential nodes, these two nodes can be combined into a supernode. Example:

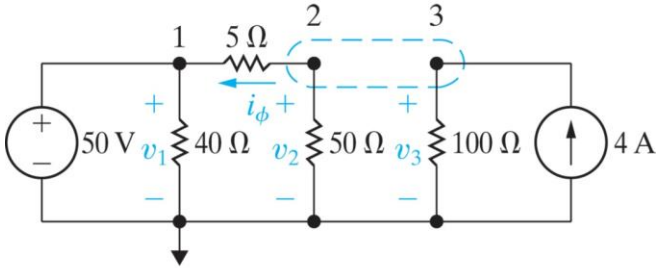


- The circuit contains four essential nodes, so we anticipate writing three node-voltage equations.
- However, two essential nodes are connected by an independent voltage source, and two other essential nodes are connected by a current-controlled dependent voltage source. Hence, there actually is only one unknown node voltage.
- We cannot express the current in the dependent voltage source branch as a function of the node voltages v_2 and v_3. Therefore i is introduced for its current.

The Node-Voltage Method: Some Special Cases







- At node 2:

$$\frac{v_2 - v_1}{5} + \frac{v_2}{50} + i = 0$$

- At node 3:

$$-i + \frac{v_3}{100} - 4 = 0$$

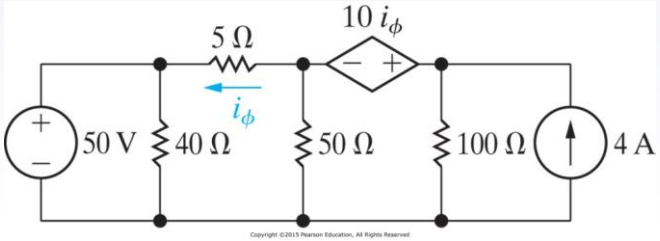
- By simply adding the two equations, we eliminate i:

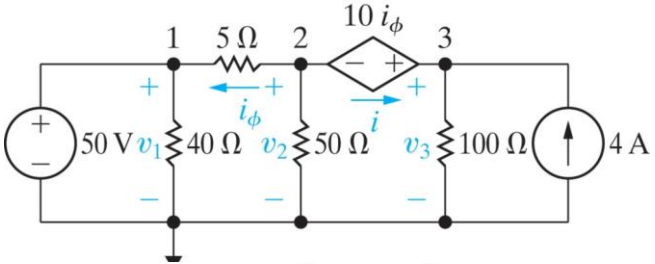
$$\frac{v_2 - v_1}{5} + \frac{v_2}{50} + \frac{v_3}{100} - 4 = 0$$

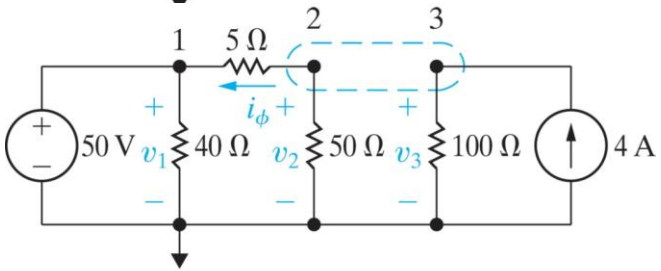
- The last equation can be written directly by considering the two essential nodes with a voltage source between them as a single node. Applying KCL:

$$\frac{v_2 - v_1}{5} + \frac{v_2}{50} + \frac{v_3}{100} - 4 = 0 \text{ (which is the same)}$$

The Node-Voltage Method: Some Special Cases







- The next step is to reduce the expression to a single unknown node voltage. First, we eliminate v_1 from the equation since we know its value: $v_1 = 50\ V$.
- Next we express v_3 as a function of v_2:

$$v_3 = v_2 + 10 i_\phi$$

- We now express the current controlling the dependent voltage source as a function of the node voltages:

$$i_\phi = \frac{v_2 - 50}{5}$$

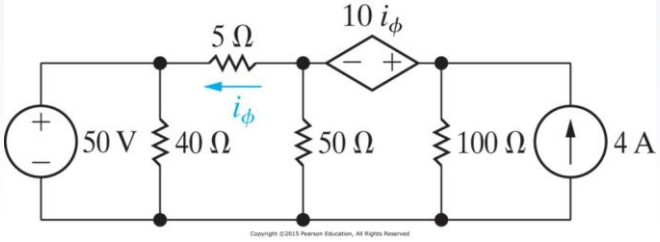
- Using all the relations in the main equation for node 2:

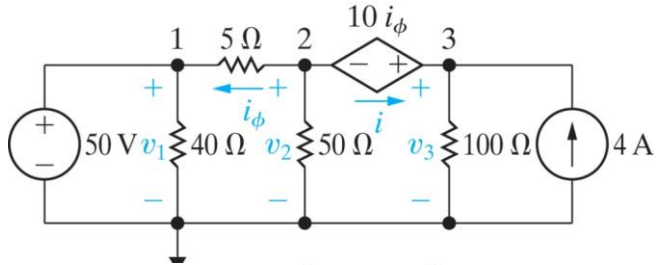
$$v_2\left(\frac{1}{50} + \frac{1}{5} + \frac{1}{100} + \frac{10}{500}\right) = 10 + 4 + 1$$

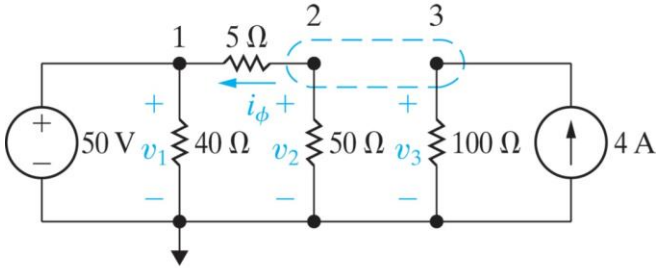
$$v_2(0.25) = 15 \qquad v_2 = 60\ V$$

The Node-Voltage Method: Some Special Cases

4







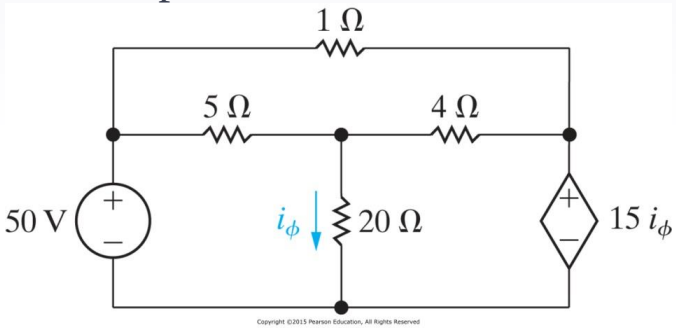
$$i_\phi = \frac{60 - 50}{5} = 2\ A$$

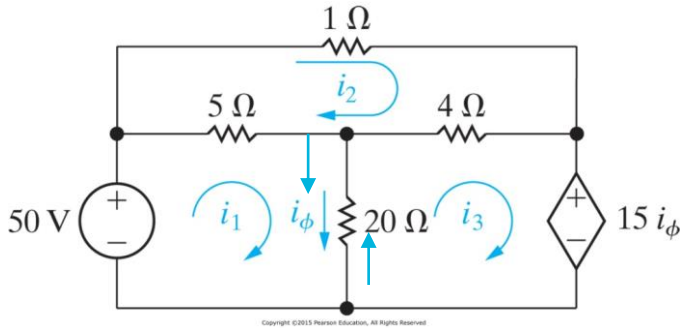
$$v_3 = 60 + 20 = 80\ V$$

The Mesh-Current Method and Dependent Sources

6

- In case there are **dependent sources** in the circuit, the constraint equations for these sources **need to be considered** as well.
- Example: Use the mesh-current method to determined the **power dissipated** in the 4 Ω resistor.





- The circuit has **6 essential branches** where the current is unknown and **4 essential nodes**. $[6-(4-1)]=\mathbf{3}$ **mesh-current equations** should be written.

For mesh 1: $-50+5(i_1-i_2)+20(i_1-i_3)=0$

For mesh 2: $5(i_2-i_1)+1i_2+4(i_2-i_3)=0$

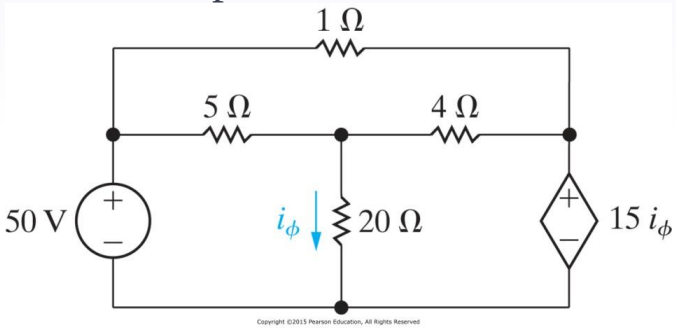
For mesh 3: $20(i_3-i_1)+4(i_3-i_2)+15i_\phi=0$

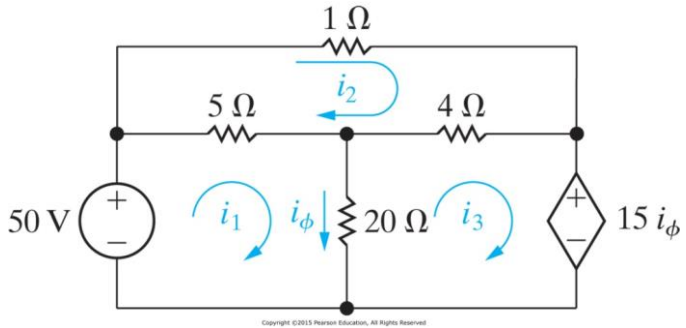
- Expressing the **branch current** controlling the dependent voltage source **in terms of the mesh currents**:

$$i_\phi = i_1 - i_3$$

The Mesh-Current Method and Dependent Sources

- In case there are **dependent sources** in the circuit, the constraint equations for these sources **need to be considered** as well.
- Example:





- **Organizing** the equations and **substituting** the last equation into the equation for mesh 3:

For mesh 1: $25i_1 - 5i_2 - 20i_3 = 50$

For mesh 2: $-5i_1 + 10i_2 - 4i_3 = 0$

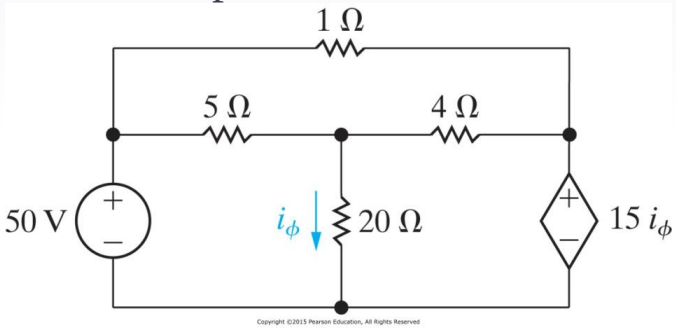
For mesh 3: $-5i_1 - 4i_2 + 9i_3 = 0$

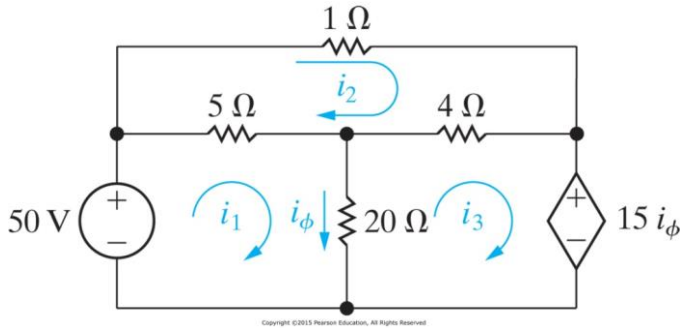
- **Subtracting** the equation for mesh 3 from the equation for mesh 2:

$$\begin{array}{r} -5i_1 + 10i_2 - 4i_3 = 0 \\ 5i_1 + 4i_2 - 9i_3 = 0 \\ \hline 14i_2 - 13i_3 = 0 \end{array}$$

The Mesh-Current Method and Dependent Sources

- In case there are **dependent sources** in the circuit, the constraint equations for these sources **need to be considered** as well.
- Example:





- **Multiplying** the equation for mesh 2 **with 5** and then **add to** the equation for mesh 1:

$$25i_1 - 5i_2 - 20i_3 = 50$$
$$-25i_1 + 50i_2 - 20i_3 = 0$$
$$45i_2 - 40i_3 = 50 \quad (dividing\ all\ by\ 5)$$
$$9i_2 - 8i_3 = 10$$

- We derived **two equations with two unknows**:

$$14i_2 - 13i_3 = 0$$
$$9i_2 - 8i_3 = 10$$
$$i_2 = 26\ A \qquad i_3 = 28\ A$$
$$p_{4\Omega} = (i_3 - i_2)^2(4) = (2)^2(4) = 16\ \text{W}$$

7

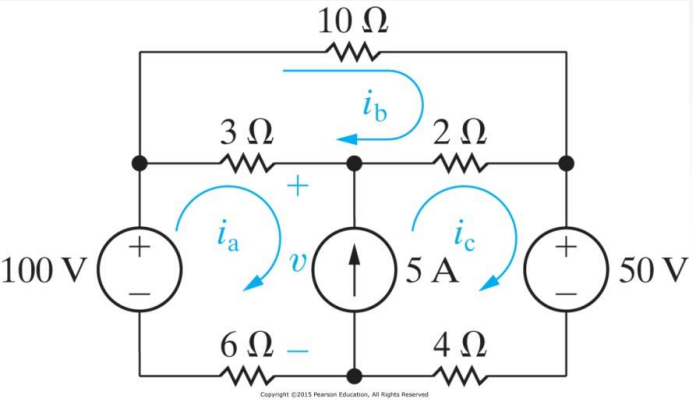
The Mesh-Current Method: Some Special Cases

- When a branch includes **a current source** the mesh-current method requires **additional manipulations**.
- In case the branch containing the current source **belongs to one mesh**, only the method is **simplified** since **the mesh current is equal to the source current**.
- In case the branch containing the current source **belongs to two meshes**, these two meshes can be **combined** into a **supermesh**.

The Mesh-Current Method: Some Special Cases

7

- Example:



- We define **mesh currents** as well as **the voltage across the 5 A current source**, to aid the discussion.
- There are **5 essential branches**, where the current is unknown and **4 essential nodes**. Therefore, we need to write $[5 - (4 - 1)] = 2$ **mesh-current equations** to solve the circuit.
- The **presence of the current source** reduces the 2 unknown mesh currents to two such currents, because it constrains the difference between i_a and i_c to equal 5 A. Hence, if we know i_a, we know i_c and vice versa.

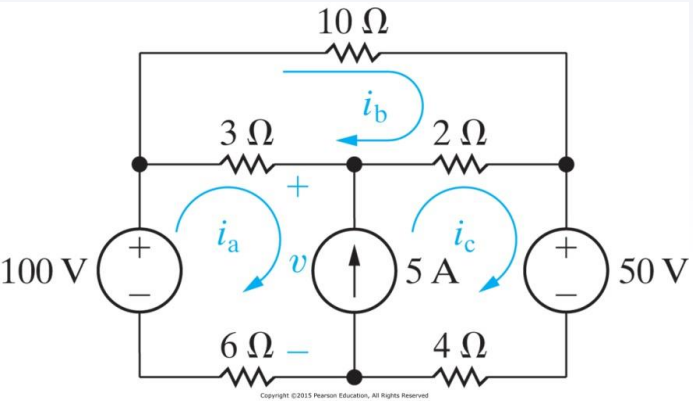
For mesh a: $-100 + 3(i_a - i_b) + v + 6i_a = 0$

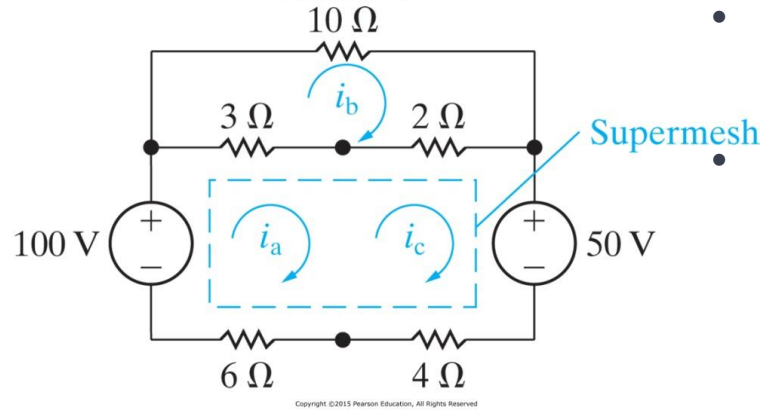
For mesh c: $-v + 2(i_c - i_b) + 50 + 4i_c = 0$

The Mesh-Current Method: Some Special Cases

7

- Example:





- Taking the sum of the equations to **eliminate** v:

$$9i_a - 5i_b + 6i_c = 50$$

- **The same equation could be directly derived** by simply **avoiding the branch with the current source** and consider a **supermesh**, when writing the equations:

$$-100 + 3(i_a - i_b) + 2(i_c - i_b) + 50 + 4i_c + 6i_a = 0$$

$$9i_a - 5i_b + 6i_c = 50$$

- **Summing** the voltages around mesh b gives

$$3(i_b - i_a) + 10i_b + 2(i_b - i_c) = 0$$

- **Reducing** both equations to two equations with two unknowns using the relation:

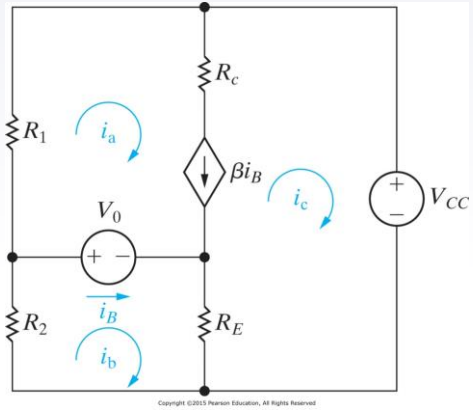
$$i_a = i_c - 5$$

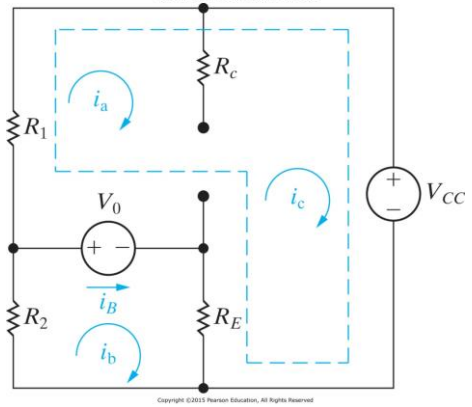
$$i_a = 1.75\ A \qquad i_b = 1.25\ A \qquad i_c = 6.75\ A$$

The Mesh-Current Method: Some Special Cases

7

- Example:





- There are **4 essential nodes** and **5 essential branches** where the current is unknown. Therefore, $[5-(4-1)]=2$ **mesh-current equations** are needed to solve the circuit.
- The **dependent current source forces a constraint** between mesh currents i_a and i_c, so we have only two unknown mesh currents.
- Considering a **supermesh**:

$$R_1 i_a + V_{CC} + R_E(i_c - i_b) - V_o = 0$$

- The mesh b equation is:

$$R_2 i_b + V_o + R_E(i_b - i_c) = 0$$

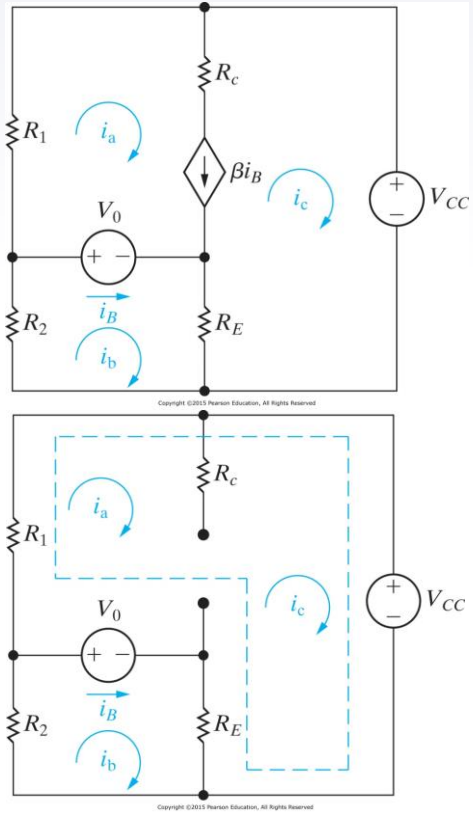
- The **constraint** imposed by the dependent current source:

$$\beta\, i_B = i_a - i_c$$

$$i_B = i_b - i_a$$

The Mesh-Current Method: Some Special Cases

- Example:



$$i_c = (1+\beta)i_a - \beta i_b$$

- **Eliminating** i_c in the mesh equations:

$$[R_1 + (1+\beta)R_E]i_a - (1+\beta)R_E i_b = V_o - V_{CC}$$

$$-(1+\beta)R_E i_a + [R_2 + (1+\beta)R_E]i_b = -V_o$$

- **Leaving** i_a and i_b alone by eliminating the other:

$$i_a = \frac{V_o R_2 - V_{CC}R_2 - V_{CC}(1+\beta)R_E}{R_1R_2 + (1+\beta)R_E(R_1+R_2)}$$

$$i_b = \frac{-V_o R_1 - (1+\beta)R_E V_{CC}}{R_1R_2 + (1+\beta)R_E(R_1+R_2)}$$

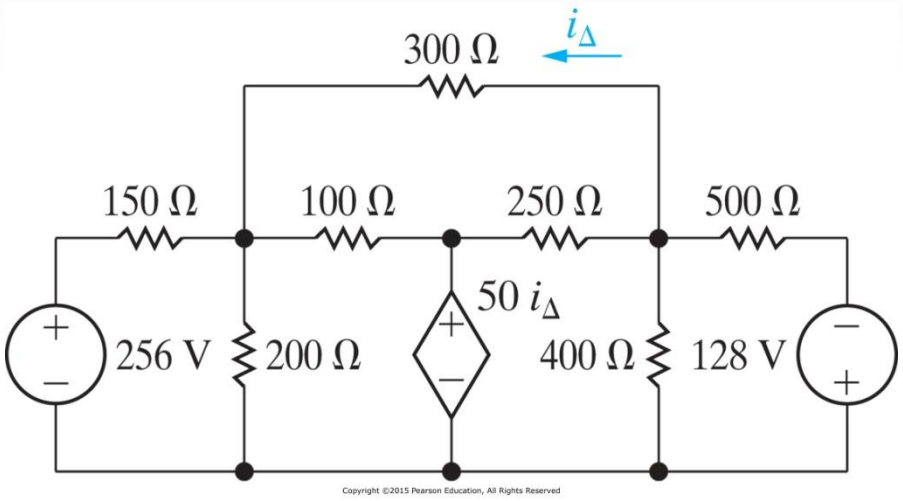
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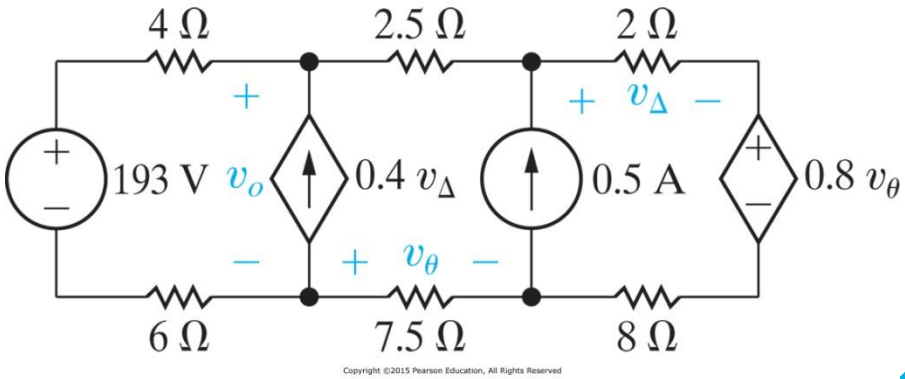
The Node-Voltage Method Versus The Mesh-Current Method

- There is **no clear answer of** when one of the methods is **preferred over the other**.
- Usually one can **try to choose** the method which **results in fewer number of equations**.
- Existence of **supernodes or supermeshes** may also be taken into account – since **they can decrease** the number of equations.
- It might be better to **spend some time to think** and consider these issues **before starting to solve** a given circuit.

The Node-Voltage Method Versus The Mesh-Current Method

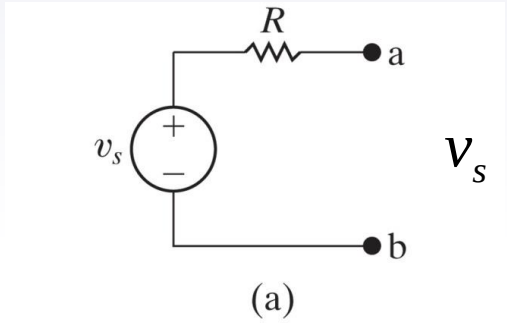
- **Two** examples:



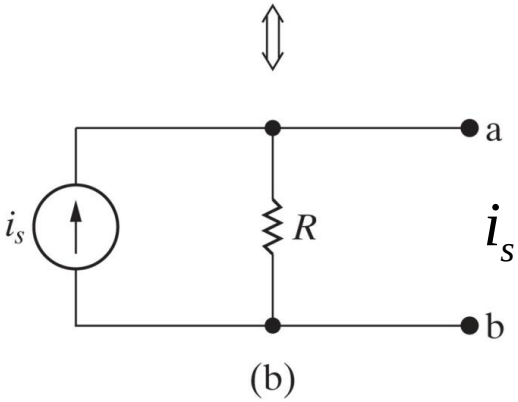


9 Source Transformations

- Source transformations can be **used to simplify** circuits during analysis.
- A source transformation **allows a voltage source in series with a resistor** to be **replaced with a current source in parallel with the same resistor** and **vice versa**.
- In order for these two circuits **to be equivalent**, we need the **current-voltage characteristics** at the a-b **terminals** of both circuits **to be the same**.
- The **polarity of v_s** and the **direction of i_s** must be as in the figure **for the equivalence to hold**.
- If the **polarity** of v_s **changes**, the **direction** of i_s **must change** as well.



$$v_s = Ri_s$$

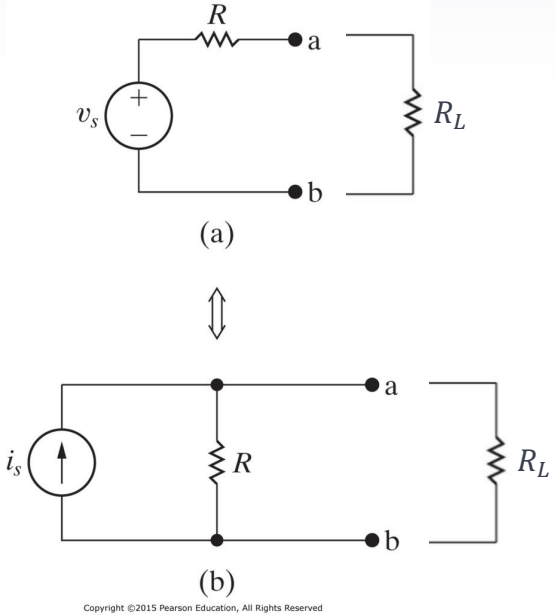


$$i_s = \frac{v_s}{R}$$

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Source Transformations

- Equivalence is **achieved** if any load resistor **experiences the same current flow**, and thus **the same voltage drop** when connected to the nodes a and b.



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$$i_L = \frac{v_s}{R + R_L}$$

$$v_s = R i_s$$

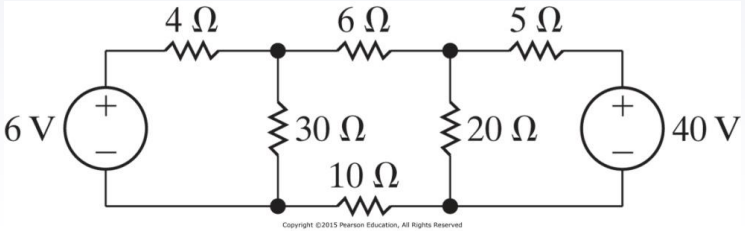
$$i_L = \frac{R}{R + R_L} i_s$$

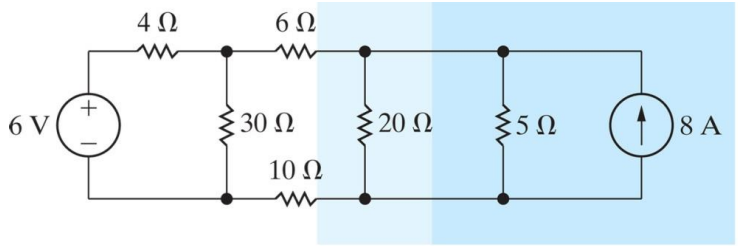
$$i_s = \frac{v_s}{R}$$

Source Transformations

9

- Example-1: Find the **power associated with** the **6 V source** and state whether it is **absorbing or delivering** that power.





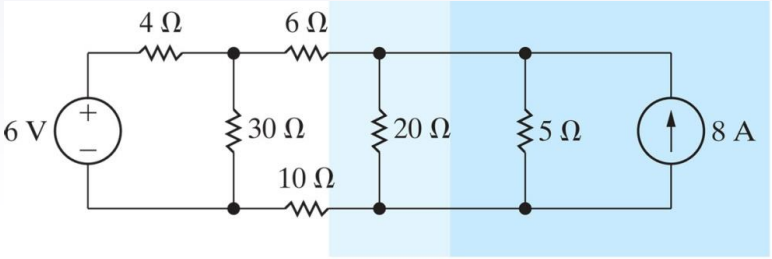
(a) First step

- The circuit has **4 essential nodes** and **6 essential branches** where the **current is unknown**. Thus we can find the current in the branch containing the 6 V source by solving, **either** (4-1)=3 node-voltage equations, or [6-(4-1)]=3 mesh-current equations.
- We can simplify the circuit by **using source transformations**. We must **reduce** the circuit in a way that **preserves the identity** of the **branch with the 6 V source**.
- We have **no reason to preserve** the **identitiy of the branch containing the 40 V source**.
- We can **transform** the 40 V source in series with the 5 Ω resistor into an 8 A current source in parallel with a 5 Ω resistor.

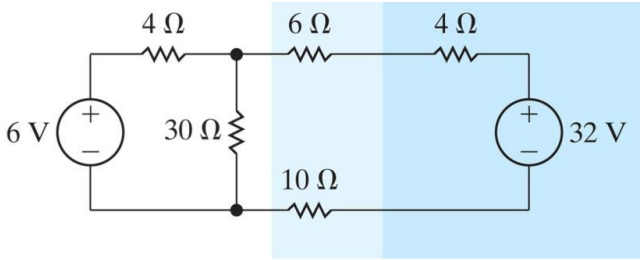
Source Transformations

9

- Example-1: Find the power associated with the 6 V source and state whether it is absorbing or delivering that power.



(a) First step



(b) Second step

- Next, we can **replace** the parallel combination of the 20 Ω and 5 Ω resistors with a 4 Ω resistor (equivalent of two parallel resistors).

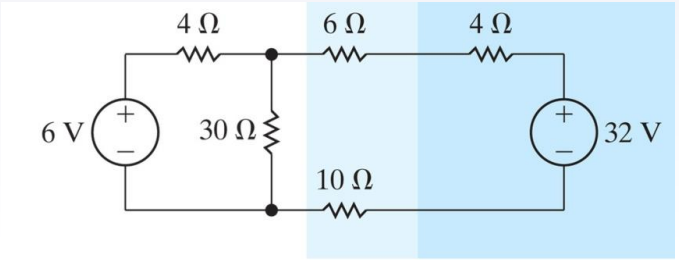
$$R_{eq1} = \frac{(20)(5)}{20+5} = 4\ \Omega$$

- This 4 Ω resistor is in parallel with the 8A source and therefore it can be **replaced** with a 32 V source in series with a 4 Ω resistor.

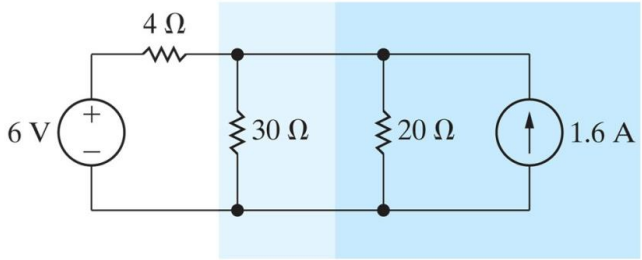
Source Transformations

9

- Example-1: Find the power associated with the 6 V source and state whether it is absorbing or delivering that power.



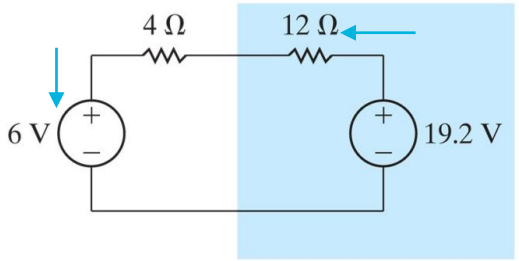
(b) Second step



(c) Third step

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- This 32 V source in series with a 20 Ω of resistance and, hence, can be **replaced** by a current source of 1.6 A in parallel with 20 Ω.
- The 20 Ω and 30 Ω parallel resistors can be **reduced** to a single 12 Ω resistor. The parallel combination of the 1.6 A current source and the 12 Ω resistor **transforms** into a voltage source of 19.2 V in series with 12 Ω.



(d) Fourth step

- The current in the direction of the voltage drop is:

$$p_{6V} = \left(\frac{19.2-6}{16}\right)(6)=4.95 \text{ W}$$

The 6 V voltage source is **absorbing power.**

served

End of the Lesson (30.10.23)

Any questions?

New supplementary videos are added to the playlist on Youtube:

Supplementary videos: Enter Youtube-> Search for ‘Alparslan Zehir’ -> Go to Playlists -> Watch ‘Electrical Circuits’ playlist